Date: ________ Day: ________

Backward Elimination:

Backward elimination is a stepwise model ~~too~~ specification technique in regression where you start with all candidate predictors and remove them one by one, based on lack of statistical significance, aiming to retain only meaningful variables.

STEPS

STEP 1: Start with full model

- Fit the complete regression model including all predictors.

Calculate

$R(\beta_1, \beta_2, \beta_3, \beta_4) = \Sigma(\hat{y} - \bar{y})^2 = SSR_{full}$

and

$$S^2 = \frac{SSE}{n-K-1}$$

STEP 2: Try Removing Each Variable (one at a time)

- For each predictor x_i in the model:

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Remove x_i, fit the reduced model

Calculate:

- $SSR_{(reduced)}^{(i)}$: SSR without x_i
- $S^2_{(full)}$ remains same.

$R\left(\beta_j \,\middle|\, \begin{matrix}\text{variables} \\ \text{already} \\ \text{in model}\end{matrix}\right) = SSR(\text{model with current} + x_i) - SSR\left(\begin{matrix}\text{current} \\ \text{model}\end{matrix}\right)$

STEP 3: Identify least significant variable.

- Find the variable x_i, whose removal causes lowest SSR and do hypothesis testing

Hypothesis:

$$H_0 : \beta_i = 0$$
$$H_1 : \beta_i \neq 0$$

Test stats:

$$F_0 = \frac{SSR}{S^2}$$

Critical Region:

$$F_\alpha (V_1, V_2)$$

$V_1 = 1$

$V_2 = N - K - 1$

N = number of obs

K = number of predictors involved.

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if $F > F_{\alpha}(v_1, v_2)$

Reject H_0.

STEP 4: Refit and Repeat.

calculate SSR_{Full}, S^2_{new}

Go back to step 2 with the updated model.

Stopping Criteria:

Continue this process until all variables left in the model are significant based on the F-test.

BACKWARD ELIMINATION:

Table 12.8: Data Relating to Infant Length*

Infant Length, y (cm)	Age, x_1 (days)	Length at Birth, x_2 (cm)	Weight at Birth, x_3 (kg)	Chest Size at Birth, x_4 (cm)
57.5	78	48.2	2.75	29.5
52.8	69	45.5	2.15	26.3
61.3	77	46.3	4.41	32.2
67.0	88	49.0	5.52	36.5
53.5	67	43.0	3.21	27.2
62.7	80	48.0	4.32	27.7
56.2	74	48.0	2.31	28.3
68.5	94	53.0	4.30	30.3
69.2	102	58.0	3.71	28.7

*Data analyzed by the Statistical Consulting Center, Virginia Tech, Blacksburg, Virginia.

Stage 1:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3 + \hat{\beta}_4 x_4$$

$$R(\beta_1, \beta_2, \beta_3, \beta_4) = 318.2744$$

$$s^2 = 0.7414.$$

Remove x_1,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3 + \hat{\beta}_4 x_4$$

$$R(\beta_1 | \beta_2, \beta_3, \beta_4) - R(\beta_1, \beta_2, \beta_3, \beta_4) - R(\beta_2, \beta_3, \beta_4)$$

$$= 0.0644$$

Remove x_2,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_3 x_3 + \hat{\beta}_4 x_4$$

$$R(\beta_2 | \beta_1, \beta_3, \beta_4) = R(\beta_1, \beta_2, \beta_3, \beta_4) - R(\beta_1, \beta_3, \beta_4)$$

$$= 0.6334$$

Remove x_3.

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_4 x_4$$

$$R(\beta_3 | \beta_1, \beta_2, \beta_4) = R(\beta_1, \beta_2, \beta_3, \beta_4) - R(\beta_1, \beta_2, \beta_4)$$

$$= 6.2523$$

Remove x_4,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3$$

$$R(\beta_4 | \beta_1, \beta_2, \beta_3) = R(\beta_1, \beta_2, \beta_3, \beta_4) - R(\beta_1, \beta_2, \beta_3)$$

$$= 0.0241$$

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By removing x_4, we get smallest SSR, so perform significance test on x_4.

Hypothesis

$H_0 : \beta_4 = 0$

$H_1 : \beta_4 \neq 0$

Test stats:

$$F = \frac{R(\beta_4 | \beta_1, \beta_2, \beta_3)}{S^2}$$

$$= \frac{0.0241}{0.7414}$$

$$= 0.0325.$$

Critical Region:

$F_\alpha (v_1, v_2)$

$v_1 = 1$

$v_2 = N-K-1$

$= 9-4-1$

$= 4$

$f_{0.05}(1,4) = 7.71$

Decision:

Accept H_0

Eliminate x_4.

Date: ______ Day: ______

Stage 2: x_4 Removed

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3$$

$R(\beta_1, \beta_2, \beta_3) = 318.2503$

$S^2 = 0.5979$

Remove x_1,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3$$

$$R(\beta_1 | \beta_2, \beta_3) = R(\beta_1, \beta_2, \beta_3) - R(\beta_2, \beta_3)$$
$$= 0.0467$$

Remove x_2,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_3 x_3$$

$$R(\beta_2 | \beta_1, \beta_3) = R(\beta_1, \beta_2, \beta_3) - R(\beta_1, \beta_3)$$
$$= 0.79$$

Remove x_3,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

$$R(\beta_3 | \beta_1, \beta_2) = R(\beta_1, \beta_2, \beta_3) - R(\beta_1, \beta_2)$$
$$= 6.233$$

By removing x_1, we get smallest SSR, so perform hypothesis testing for x_1.

Date: Day:

Hypothesis:

$H_0 : \beta_1 = 0$

$H_1 : \beta_1 \neq 0$

~~Critical R~~ Test Stats

$$F = \frac{R(\beta_1 | \beta_2, \beta_3)}{S^2}$$

$= 0.0781$

Critical Region:

$f_\alpha(v_1, v_2)$ $v_2 = 9-3-1$

$f_{0.05}(1, 5) = 6.61$

Decision:

Accept H_0.

x_1 eliminated.

Stage 3:

$$\hat{y} = \hat{\beta_0} + \hat{\beta_2} x_2 + \hat{\beta_3} x_3$$

$R(\beta_2, \beta_3) = 318.2036$

$S^2 = 0.5061$

Remove x_2.

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_3 x_3$$

$R(\beta_2 | \beta_3) = R(\beta_2, \beta_3) - R(\beta_3)$

$= 132.0971$

Date:______ Day:______

Remove x_3,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_2 x_2$$

$$R(\beta_3 | \beta_2) = R(\beta_2, \beta_3) - R(\beta_2)$$

$$= 102.9023.$$

By removing x_3, SSR is smallest so perform significance test for x_3.

Hypothesis:

$H_0 : \beta_3 = 0$

$H_1 : \beta_3 \neq 0$

Test stats:

$$F = \frac{R(\beta_3 | \beta_2)}{S^2}$$

$$= 203.3240$$

Critical Region:

$F_{\alpha\, 0.05}(1, 6) = 5.99.$

Decision:

x_3 is significant, keep x_3

H_0 Reject

Final model,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3.$$